

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Technology (B.Tech)
Sem - II University Examination, May-2010

B.Tech(CE/EE/IT)
ME-101 Engineering Graphics

Date: 22.05.2010, Saturday

Time: 12:30 p.m. to 03:30 p.m.

Maximum Marks: 70

Instructions:

1. Figures to the right indicate **full** marks.
2. Make suitable assumptions and draw neat figures wherever if required.

SECTION- I

Q1 (a) A 3.2 cm long line represents a length of 4 meters. Extend this line to measure lengths up to 20 metres and show on it units of meters and 5 meters. Also show the length of 13 meters on this line. 04

(b) Two lines AB and CD are 100 mm and 80 mm long respectively and bisect each other at right angle. Draw the curve passing through all end points of these lines. Use (1) Oblong Method to complete half portion, and (2) Concentric Circle Method to complete half portion of the curve. Name the complete curve 06

Q2 The top view of 75 mm long line AB measures 50 mm. A is 50 mm in front of VP and 15 mm below HP. B is 15 mm in front of VP and is above the HP. Draw the front view of AB and find its all the inclinations with the reference planes. 10

OR

Q2 The end projectors of line PQ are 65 mm apart. P is 25 mm behind the VP and 30 mm below the HP. The point Q is 40 mm above the HP and 15 mm in front of VP. Draw its projections and find its true length, all the inclinations of line PQ with reference planes. 10

Q3 Fig. 1 shows pictorial view of an object. Draw (1) Front view in the direction 'X', (2) Left hand side view and (3) Top view. 15

OR

Q3 Fig. 2 shows pictorial view of an object. By Using first angle projection method, Draw (1) Half Sectional Front view in the direction 'X', (2) Top view. 15

SECTION - II

Q4 Draw Isometric View for Figure-3 12

Q 5 A vertical square prism, base 50 mm side, has a face inclined at 30° to the V.P. It has a hole of 65 mm diameter drilled through it. The centre line of the hole is parallel the H.P. and the V.P. and is 5 mm away from the axis of the prism. Draw the projection of the prism. Also draw the development of surfaces of remaining part of solid. 18

OR

Q 5 Attempt any Two 18

(i) Draw the projections of a regular pentagon of 40 mm side, having its surface inclined at 30° to the HP and a side parallel to the HP and inclined at an angle of 60° to the VP

(ii) A square pyramid of base side 30 mm and height 60 mm is resting on HP on its base with one of the edge of base perpendicular to VP. It is cut by a section plane perpendicular to VP and inclined to HP at 45° and passing through a point 20 mm from apex on the axis. Draw the projections. Draw the development of surfaces of cut portion of pyramid having base in it.

(iii) Draw the projection of a cube of 30 mm long edges resting on the ground on one of its corners with a solid diagonal perpendicular to V.P.

Q6 Give the difference between first angle and third angle projection method with symbol 05

